

Final Project Draft

Abstract:

Carbon dioxide emissions play a crucial role in understanding the relationship between economic development, energy consumption, and climate change. This research is motivated by the broader challenge of climate change. In this paper, we will explore and analyze data from *Our World in Data CO2* to study how global emissions vary across countries and over time while exploring how economic and energy-related factors are associated with CO2 emissions. Due to the nature of the dataset, the analysis also considers the importance of time-series modeling and evaluation. This report is intended to provide a structured, data-driven analysis of global emission trends and their relationship to broader economic and energy indicators.

Keywords: Climate change, CO2 emissions, Machine Learning, Public Health

Introduction:

Human emissions of greenhouse gases are the primary drive of climate change today. As economies, populations, and energy demand grows, emissions will grow as well. However, the strength and structure of these relationships will vary across countries and over time. Understanding these patterns is important for social and science advancement. From a scientific perspective, researchers want to better understand how economic development, population growth, and energy consumption connect to emissions. From a more societal perspective, these relationships are connected to climate policy, sustainable actions, and environmental risk on a more global scale.

This problem is challenging to researchers as emissions are influenced by many interconnected factors, including industrialization, energy infrastructure, economic scale, and different policies around the world. The data is also collected over time meaning the observations are also known as time series data which must be counted for during the analysis. Ignoring the analysis of a time-series can produce misleading results when evaluating predictive models.

This research will explore and analyze country-level data from the *Our World in Data CO2* and Greenhouse Gas Emissions dataset, focusing on observations from 1990 and onward. The goals of this analysis are to explore global and country-level trends in CO2 emissions, examine how emissions relate to GDP, populations, and primary energy consumption, and evaluate predictive models for CO2 emissions from these covariates. Due to the nature of this dataset, the time series structure will be considered in the data splitting and model evaluation process.

Data Collection:

The data used in this project comes from the Our World in Data CO2 and the Greenhouse Gas emissions dataset, compiled of country-level environmental, demographic, and economic indicators from multiple international sources into a standardized longitudinal dataset. Each observation corresponds to a country year to allow the data to capture how emissions and other related variables change over time with the countries.

Not all variables in the dataset were used. Only the variables most relevant were selected to the scientific question of interest, including country, year, iso_code, co2, co2_per_capita, gdp, population, and primary_energy_consumption. Total CO2 emissions (co2) serve as the primary outcome of interest because they directly represent economic output. Population captures demographic scale while primary energy consumptions represents energy demand. These variables are relevant because they represent major drive that are often linked to national emission levels. A useful feature of the dataset is that it includes repeated yearly observations, making it easier to study emission trends over time rather than relying on a single cross section. There are also limitations to using this dataset. The data is observational so the results should not be casually interpreted. The dataset also included both countries and aggregate regional entities so filtering was done to make the data more aligned with the focus of the research. Some important variables also contained missing values which could affect the modeling sample size.

Data Preprocessing:

Several preprocessing steps were performed to create the dataset used for analysis. A subset of the variables related to emissions, economic output, population, and energy consumption was selected. The data was then restricted to years 1990 and later to focus on recent observations and trends. Missing iso_code values were removed to focus on country-level units that are easily identifiable rather than broader regional or economic observations.

After the cleaning, the main variables were examined to find any missing values. Some key predictors had missing values so two related datasets odf_clean and odf_model were created for descriptive exploration and refined observations. The second dataset, odf_model, was created by dropping observations with missing values in the relevant variables so that all fitted models are trained on the same variables and such.

Exploratory plots were created using ggplot2 and informed preprocessing decisions. Plotting global CO2 emissions over time showed a strong upward trend.

Methods:

The first step was EDA. Moving forward, unsupervised learning methods, and predictive modeling is implemented. EDA mainly focused on understanding global trends, country differences, and the relationship between GDP and emissions.

Draft part after...

I plan to add

- Visual Plots with nice colors (ggplot, pca analysis, etc) to their respective sections
- Plots of PCA
- Time-based data splitting
- Baseline model
- Full log linear regression
- Reduced log-linear regression

Results section

Discussions section which include revised limitations, future implications and improvements and what this means for the field of carbon emissions, climate change, and machine learning

Citations